

Ashvin Ranjan

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Education

University of Washington | Expected June 2027

GPA: 3.98/4.0

Bachelor of Science in Computer Science, Minor in Linguistics

- Courses: Introduction to Computer Programming I, II, & III, Hardware/Software Interface, Foundations of Computing I, Data Structures & Algorithms, Software Design & Implementation, Shallow Processing Techniques for NLP (Master's course).
 - Complete by June 2026: Foundations of Computing II, Advanced Statistical Methods in NLP (Master's course), Systems Programming, Operating Systems.
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Skills

Languages: Typescript, Javascript, Python, Rust, Go, Java, C#

Next.js and React Frontend development | Typescript, Rust, and Go backend development

Research in Linguistics and Computer Science

Work

CLMBR | Prof. Steinert-Threlkeld's Computational Linguistics research lab.

Undergraduate Researcher

November 2024 - Present

- First author on research using a custom **numpy** and **pandas**-based **python** library to analyze Information Bottleneck systems.
- Leading research on regular expression matching at **terabyte scale** utilizing custom **CUDA GPU kernels** and algorithms.
- Worked on pretraining and fine-tuning for various **Hugging Face transformers** across a range of languages and forms of data.

Inspirogram | A Computer Science education non-profit.

Software Engineering Intern

May 2022 - January 2023

- Volunteered to help develop the website and write courses for the non-profit, which would be used to teach students in Tanzania.
- Utilized **Next.js** to develop the website. Wrote courses on **Unity** and **React.js** software development and made presentations.

Outreach | A marketing and outreach company.

Software Engineering Intern

July 2022 - August 2022

- Utilized **Go** to develop a backend and command line interface to create company GitHub repositories with custom templates.
- Translated Ruby code into **Go** and created a custom YAML specification to allow for the creation of custom repository templates.

Software Engineering Intern

June 2021 - August 2021

- Collaborated with the calendaring experiences team in order to solve bugs in the calendar of the **Next.js**-based company website.
- Created outline for additional sections of the company website involving image processing and uploading by users using **Next.js**.

Backbone | A mobile gaming peripherals company.

Software Engineering Intern

March 2020 - August 2020

- Created an iOS application with **Swift** to send live stream data to a computer via USB to an OBS plugin for recording or streams.
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Publications

Preprints

[Ashvin Ranjan](#) & Shane Steinert-Threlkeld. (2026). When Efficient Communication Explains Convexity. *arXiv*.

<https://arxiv.org/pdf/2602.02821>

Projects

EMP | github.com/Ashvin-Ranjan/EMP

September 2021 - January 2022

- Designed a bytecode schema for data packaging that is **12%** more efficient than Non-Binary Tags and Javascript Object Notation.
- Developed a library and command line interface for the schema in **Rust**, published on crates.io with more than 7,600 downloads.

Disadus | github.com/disadus

August 2021 - April 2022

- Produced a learning management system with other students reaching over 300 registered users and over 10 student communities.
- Architected and developed the **Next.js** website and **Typescript** backend which integrated MongoDB, node mailer, and **Firebase**.